The Closure Residual: Nonlinear Remainder Beyond the Witten-Hamiltonian Regime

Lee Smart

Institute of Vibrational Field Dynamics

contact@vibrationalfielddynamics.org

@vfd_org

April 2026

Status: Preprint (not peer-reviewed)

Abstract

Paper XVIII derived an exact nonlinear closure-Nelson wave equation and showed that the nonlinear residual vanishes exactly at the Paper XIV equilibrium, so that the wave equation reduces to the Schrödinger equation with Witten Hamiltonian H_{W}; for perturbations around equilibrium, this linear Schrödinger dynamics persists to leading order. The remaining discrepancy away from equilibrium is a state-dependent nonlinear term with coefficient σ^2 in the Schrödinger-normalised form.

The present paper studies this *closure residual* directly. Rewriting the dynamics in equilibrium-centred normal form $i\partial_t\Psi = H_{\mathrm{W}}\Psi + \delta U_{\mathrm{rel}}[\Psi]\Psi$, the residual is shown to vanish exactly at the Paper XIV equilibrium and to measure curvature deviation of the probability amplitude from equilibrium structure. Structural properties are derived: global phase invariance, modulus-only dependence, locality, and near-equilibrium smallness. The full nonlinear equation is shown to conserve the L^2 norm exactly, confirming that probability is preserved even in the nonlinear regime. A gauge analysis establishes that the residual cannot be removed by phase redefinition or linear unitary transformation, and that amplitude-based absorption changes the physical content of the theory.

A dimensionless residual-strength parameter Ξ is defined, providing a regime classifier for the dynamics: a linear Schrödinger regime ($\Xi \ll 1$), a mixed regime ($\Xi \sim 1$), and a nonlinear closure-dominated regime ($\Xi \gg 1$). A quadratic worked example demonstrates each regime explicitly.

The paper does not interpret the residual as a modification of quantum mechanics. It establishes it as a mathematically well-defined object — the closure-sector remainder after linear quantum behaviour has been recovered — whose physical interpretation is deferred to subsequent work.

1 Introduction

Paper XVIII [Smart, 2026b] established that forward/backward pairing of closure stochastic processes produces a nonlinear Schrödinger-type wave equation:

$$i\sigma^2\partial_t\Psi = -\frac{\sigma^4}{2}\Delta\Psi + \left(-\sigma^2 V_{\mathrm{W}} + \sigma^4\frac{\Delta|\Psi|}{|\Psi|}\right)\Psi \tag{1}$$

where $V_{\mathrm{W}} = |\nabla\mathcal{F}|^2/(2\sigma^2) - \Delta\mathcal{F}/2$ is the Witten potential. At the Paper XIV [Smart, 2026a] equilibrium and to leading order near equilibrium, this reduces to the linear Schrödinger equation $i\partial_t\Psi = H_{\mathrm{W}}\Psi$ with Witten Hamiltonian $H_{\mathrm{W}} = -(\sigma^2/2)\Delta + V_{\mathrm{W}}$.

The sole remaining obstruction to global linear Schrödinger dynamics is the state-dependent term $\sigma^4\Delta|\Psi|/|\Psi|$ (or equivalently $\sigma^2\Delta|\Psi|/|\Psi|$ after normalisation to Schrödinger form). Paper XVIII identified this as a structurally derived residual rather than a defect of the construction. The present paper takes the next step: it isolates, classifies, and analyses this residual mathematically.

Scope

This paper does not interpret the residual as new physics. It establishes the residual's mathematical properties so that subsequent work can determine its physical status. The paper answers: *what kind of object is the residual, and when does it matter?*

2 Equilibrium-Centred Normal Form

Define the stationary amplitude $R_{\mathrm{st}} = \sqrt{\rho_{\mathrm{st}}} \propto e^{-\mathcal{F}/\sigma^2}$, where $\rho_{\mathrm{st}} = Z^{-1}e^{-2\mathcal{F}/\sigma^2}$ is the Paper XIV equilibrium density. The key identity (derived in Paper XVIII) is:

$$\frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} = \frac{|\nabla\mathcal{F}|^2}{\sigma^4} - \frac{\Delta\mathcal{F}}{\sigma^2} = \frac{2V_{\mathrm{W}}}{\sigma^2} \tag{2}$$

Dividing (1) by σ^2 and using (2), the closure-Nelson wave equation takes the *equilibrium-centred normal form*:

$$\boxed{i\,\partial_t\Psi = H_{\mathrm{W}}\Psi + \delta U_{\mathrm{rel}}[\Psi]\,\Psi} \tag{3}$$

where the *closure residual* is:

$$\boxed{\delta U_{\mathrm{rel}}[\Psi] = \sigma^2\left(\frac{\Delta|\Psi|}{|\Psi|} - \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}}\right)} \tag{4}$$

Remark 1. *This form separates the dynamics into two components: the linear Witten-Hamiltonian evolution $H_{\mathrm{W}}\Psi$ (which is exact Schrödinger), and the nonlinear closure residual $\delta U_{\mathrm{rel}}[\Psi]\Psi$ (which vanishes at equilibrium). The residual measures how far the current amplitude profile deviates from the equilibrium curvature structure. The coefficient σ^2 results from dividing Paper XVIII's wave equation (which has coefficient σ^4 in the undivided form $i\sigma^2\partial_t\Psi = \ldots$) by σ^2 to obtain the Schrödinger-normalised form.*

3 Structural Properties of the Residual

Proposition 2 (Equilibrium vanishing)**.** *$\delta U_{\mathrm{rel}}[\Psi_{\mathrm{st}}] = 0$, where $\Psi_{\mathrm{st}} = R_{\mathrm{st}}$ is the equilibrium amplitude.*

Proof. Direct substitution: $|\Psi_{\mathrm{st}}| = R_{\mathrm{st}}$, so $\Delta|\Psi_{\mathrm{st}}|/|\Psi_{\mathrm{st}}| = \Delta R_{\mathrm{st}}/R_{\mathrm{st}}$, and the difference vanishes. □

Proposition 3 (Phase invariance)**.** *$\delta U_{\mathrm{rel}}[e^{i\theta}\Psi] = \delta U_{\mathrm{rel}}[\Psi]$ for any constant $\theta \in \mathbb{R}$. More generally, $\delta U_{\mathrm{rel}}[\Psi]$ depends only on $|\Psi|$ and its spatial derivatives, not on the phase of Ψ.*

Proof. $|e^{i\theta}\Psi| = |\Psi|$, so $\Delta|e^{i\theta}\Psi|/|e^{i\theta}\Psi| = \Delta|\Psi|/|\Psi|$. □

Proposition 4 (Locality)**.** *On any open set where $|\Psi| > 0$, $\delta U_{\mathrm{rel}}[\Psi](\psi)$ depends only on $|\Psi|$, $\nabla|\Psi|$, and $\Delta|\Psi|$ evaluated at ψ. It is a local functional of the modulus.*

Proof. On $\{|\Psi| > 0\}$: $\Delta|\Psi|/|\Psi| = \Delta \log|\Psi| + |\nabla \log|\Psi||^2$, which involves only local derivatives of $|\Psi|$. □

Proposition 5 (Near-equilibrium scaling)**.** *For $\Psi = \Psi_{\mathrm{st}}(1+\varepsilon\eta)$ with $\varepsilon \ll 1$ and η smooth:*

$$\delta U_{\mathrm{rel}}[\Psi] = O(\varepsilon) \tag{5}$$

More precisely, to leading order:

$$\delta U_{\mathrm{rel}}[\Psi] = \varepsilon\sigma^2 \left(\Delta \mathrm{Re}(\eta) + 2\nabla \log R_{\mathrm{st}} \cdot \nabla \mathrm{Re}(\eta)\right) + O(\varepsilon^2) \tag{6}$$

Proof. Write $R = |\Psi| = R_{\mathrm{st}}(1 + \varepsilon\,\mathrm{Re}(\eta)) + O(\varepsilon^2)$. Then:

$$\frac{\Delta R}{R} = \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} + \varepsilon\left(\Delta \mathrm{Re}(\eta) + 2\nabla \log R_{\mathrm{st}} \cdot \nabla \mathrm{Re}(\eta)\right) + O(\varepsilon^2) \tag{7}$$

Subtracting $\Delta R_{\mathrm{st}}/R_{\mathrm{st}}$ gives (6). □

Proposition 6 (Characterisation of vanishing)**.** *$\delta U_{\mathrm{rel}}[\Psi] = 0$ if and only if $|\Psi|$ satisfies:*

$$\frac{\Delta|\Psi|}{|\Psi|} = \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} = \frac{2V_{\mathrm{W}}}{\sigma^2} \tag{8}$$

That is, the residual vanishes precisely for states whose amplitude has the same curvature-to-amplitude ratio as the equilibrium profile.

Proof. Direct from definition (4). □

4 Conservation Laws

Proposition 7 (Norm conservation)**.** *The full nonlinear equation (3) preserves the L^2 norm:*

$$\frac{d}{dt}\int |\Psi|^2\, d\psi = 0 \tag{9}$$

Proof. Write (3) as $i\partial_t\Psi = H_{\mathrm{W}}\Psi + \delta U_{\mathrm{rel}}\Psi$. Then:

$$\begin{aligned}\frac{d}{dt}\int |\Psi|^2 &= 2\,\mathrm{Re}\int \Psi^*\,\partial_t\Psi\,d\psi \\ &= 2\,\mathrm{Re}\int \Psi^*\left(-iH_{\mathrm{W}}\Psi - i\,\delta U_{\mathrm{rel}}\Psi\right)d\psi \end{aligned} \tag{10}$$

For the Witten term: $\mathrm{Re}\int \Psi^*(-iH_{\mathrm{W}}\Psi) = -\mathrm{Im}\langle\Psi, H_{\mathrm{W}}\Psi\rangle = 0$ since H_{W} is self-adjoint and $\langle\Psi, H_{\mathrm{W}}\Psi\rangle$ is real.

For the residual term: $\mathrm{Re}\int \Psi^*(-i\,\delta U_{\mathrm{rel}}\Psi) = -\mathrm{Im}\int \delta U_{\mathrm{rel}}\,|\Psi|^2 = 0$ since $\delta U_{\mathrm{rel}}[\Psi]$ is *real-valued* (it depends only on modulus derivatives via (4)). □

Remark 8 (Physical significance)**.** *Norm conservation holds exactly for the full nonlinear equation, not just in the linearised regime. Probability is preserved even when the residual is large. This is because δU_{rel} acts as a real multiplicative potential: it rotates the phase of Ψ but does not drain or amplify its modulus. The closure-Nelson dynamics is therefore probability-conserving in all regimes.*

Remark 9 (On unitarity vs. norm conservation)**.** *The closure dynamics preserves the L^2 norm but is not generated by a skew-adjoint operator. In this sense it is norm-preserving but not unitary in the standard quantum-mechanical sense: the generator $H_{\mathrm{W}} + \delta U_{\mathrm{rel}}[\Psi]$ is self-adjoint only in the linearised regime where $\delta U_{\mathrm{rel}} \to 0$.*

Continuity equation

Since the norm is conserved, the density $\rho = |\Psi|^2$ satisfies a continuity equation. Writing $\Psi = R\,e^{iS/\sigma^2}$, the Madelung decomposition gives:

$$\partial_t\rho + \nabla\cdot(\rho\,\nabla S) = 0 \tag{11}$$

which is exact for the full nonlinear equation (inherited from Paper XVIII's Madelung pair).

Energy functional

Define the candidate energy:

$$E[\Psi] = \langle\Psi, H_{\mathrm{W}}\Psi\rangle = \int\left(\frac{\sigma^2}{2}|\nabla\Psi|^2 + V_{\mathrm{W}}|\Psi|^2\right)d\psi \tag{12}$$

Proposition 10 (Energy non-conservation)**.** *The Witten energy $E[\Psi]$ is **not** conserved by the full nonlinear dynamics. Its time derivative is:*

$$\frac{dE}{dt} = -2\,\mathrm{Im}\int \delta U_{\mathrm{rel}}[\Psi]\,\Psi^*\,H_{\mathrm{W}}\Psi\,d\psi \tag{13}$$

which vanishes at equilibrium and to leading order near equilibrium, but is generically nonzero.

Proof. $dE/dt = 2\,\mathrm{Re}\langle\partial_t\Psi, H_{\mathrm{W}}\Psi\rangle = 2\,\mathrm{Re}\langle -iH_{\mathrm{W}}\Psi - i\delta U_{\mathrm{rel}}\Psi, H_{\mathrm{W}}\Psi\rangle$. The first term gives $-2\,\mathrm{Im}\langle H_{\mathrm{W}}\Psi, H_{\mathrm{W}}\Psi\rangle = 0$ (since H_{W} is self-adjoint). The second gives $-2\,\mathrm{Im}\langle\delta U_{\mathrm{rel}}\Psi, H_{\mathrm{W}}\Psi\rangle = -2\,\mathrm{Im}\int \delta U_{\mathrm{rel}}[\Psi]\,\Psi^*\,H_{\mathrm{W}}\Psi\,d\psi$, which is generically nonzero since δU_{rel} is real but $\Psi^* H_{\mathrm{W}}\Psi$ is generally complex. □

Remark 11. *The functional $E[\Psi] = \langle \Psi, H_{\mathrm{W}}\Psi\rangle$ is the linear Witten energy — the energy of the linearised Schrödinger regime. It is not necessarily the conserved energy of the full nonlinear theory. Norm is conserved but linear Witten energy is not: the nonlinearity exchanges energy between the Witten-Hamiltonian sector and the residual sector. Whether a modified energy functional incorporating the residual exists and is conserved remains open.*

5 Gauge and Removability Analysis

We test whether the residual can be absorbed by admissible transformations.

5.1 Phase redefinition

Proposition 12 (Non-removability by phase)**.** *No transformation $\Psi \to \Psi\, e^{iG(|\Psi|)}$ with G depending only on $|\Psi|$ can remove δU_{rel}.*

Proof. Under $\Psi \to \Psi\, e^{iG}$, the modulus $|\Psi|$ is unchanged, so $\delta U_{\mathrm{rel}}[\Psi\, e^{iG}] = \delta U_{\mathrm{rel}}[\Psi]$. The residual is phase-invariant (Proposition 3). □

5.2 Linear unitary transformation

Proposition 13 (Non-removability by linear unitary)**.** *No time-independent unitary transformation $\Psi \to U\Psi$ (with U a fixed linear operator) can remove δU_{rel} from the equation.*

Proof. Under $\Psi \to U\Psi$, the equation transforms as $i\partial_t(U\Psi) = UH_{\mathrm{W}}U^{\dagger}(U\Psi) + \delta U_{\mathrm{rel}}[U\Psi](U\Psi)$. The residual $\delta U_{\mathrm{rel}}[U\Psi] = \sigma^2(\Delta|U\Psi|/|U\Psi| - \Delta R_{\mathrm{st}}/R_{\mathrm{st}})$ depends on $|U\Psi|$, which generically differs from $|\Psi|$ for non-trivial U. The residual is modified but not eliminated. □

5.3 Equilibrium-amplitude factorisation

The substitution $\Psi = R_{\mathrm{st}}\,\Phi$ (used in Paper XVI to eliminate the drift) transforms the equation. Since $|\Psi| = R_{\mathrm{st}}|\Phi|$:

$$\frac{\Delta|\Psi|}{|\Psi|} = \frac{\Delta(R_{\mathrm{st}}|\Phi|)}{R_{\mathrm{st}}|\Phi|} \tag{14}$$

Expanding: $\Delta(R_{\mathrm{st}}|\Phi|) = |\Phi|\Delta R_{\mathrm{st}} + 2\nabla R_{\mathrm{st}} \cdot \nabla|\Phi| + R_{\mathrm{st}}\Delta|\Phi|$, so:

$$\frac{\Delta|\Psi|}{|\Psi|} = \frac{\Delta R_{\mathrm{st}}}{R_{\mathrm{st}}} + \frac{2\nabla \log R_{\mathrm{st}} \cdot \nabla|\Phi|}{|\Phi|} + \frac{\Delta|\Phi|}{|\Phi|} \tag{15}$$

The residual becomes:

$$\delta U_{\mathrm{rel}} = \sigma^2 \left(\frac{2\nabla \log R_{\mathrm{st}} \cdot \nabla|\Phi|}{|\Phi|} + \frac{\Delta|\Phi|}{|\Phi|} \right) \tag{16}$$

Remark 14. *The equilibrium piece $\Delta R_{\mathrm{st}}/R_{\mathrm{st}}$ cancels exactly, but the residual is not eliminated: it is re-expressed in terms of $|\Phi|$ and its derivatives. The residual vanishes if and only if $|\Phi|$ is constant (i.e. $|\Psi| \propto R_{\mathrm{st}}$). The factorisation simplifies the residual's dependence but does not remove it.*

Table 1: Removability of the closure residual under candidate transformations.

Transformation	Effect on δU_{rel}	Removable?
Global phase $e^{i\theta}$	Invariant	No
Phase redefinition $e^{iG(\lvert\Psi\rvert)}$	Invariant	No
Linear unitary U	Modified, not eliminated	No
Equilibrium factorisation $R_{\mathrm{st}}\Phi$	Re-expressed in $\lvert\Phi\rvert$	No
Nonlinear amplitude transform	May absorb, changes theory	Open

5.4 Summary of removability

The residual is robust under all transformations that preserve the physical content of the theory. It can only be absorbed by transformations that alter the equation's structure.

Proposition 15 (Homogeneity under amplitude rescaling)**.** *For any constant $\lambda > 0$:*

$$\delta U_{\mathrm{rel}}[\lambda\Psi] = \delta U_{\mathrm{rel}}[\Psi] \tag{17}$$

Thus the residual depends on the shape of the modulus profile, not on its overall normalisation.

Proof. Since $|\lambda\Psi| = \lambda|\Psi|$, $\Delta|\lambda\Psi|/|\lambda\Psi| = \Delta|\Psi|/|\Psi|$. The equilibrium term is unchanged, so $\delta U_{\mathrm{rel}}[\lambda\Psi] = \delta U_{\mathrm{rel}}[\Psi]$. □

6 Residual Strength and Dynamical Regimes

Define the dimensionless *residual-strength parameter*:

$$\Xi[\Psi](\psi) = \frac{|\delta U_{\mathrm{rel}}[\Psi](\psi)|}{|V_{\mathrm{W}}(\psi)| + \sigma^2|\Delta\Psi/\Psi|/2} \tag{18}$$

as a pointwise ratio of the residual magnitude to the characteristic scale of the Witten-Hamiltonian terms.

Three regimes:

Table 2: Dynamical regimes classified by residual strength.

Regime	Condition	Dynamics
Linear Schrödinger	$\Xi \ll 1$	$i\partial_t\Psi \approx H_{\mathrm{W}}\Psi$
Mixed	$\Xi \sim 1$	Witten + residual compete
Nonlinear closure	$\Xi \gg 1$	Residual dominates

Remark 16. *The linear Schrödinger regime contains the equilibrium ($\Xi = 0$) and its perturbative neighbourhood. The nonlinear closure regime corresponds to configurations whose density profiles differ strongly from ρ_{st} — for instance, highly localised or rapidly oscillating amplitude profiles. The mixed regime is the transitional zone where both components contribute comparably. This pointwise quantity is defined on regions where $|\Psi| > 0$; global (integrated) versions and treatment near nodal sets are left for future work.*

7 Worked Example: Quadratic Closure Functional

Example 17 (Quadratic closure)**.** *Let $\mathcal{F}(\psi) = \frac{1}{2}k\psi^2$ in one dimension.*

Equilibrium structure

$R_{\rm st} = \mathcal{N}\, e^{-k\psi^2/(2\sigma^2)}$, $V_{\rm W} = k^2\psi^2/(2\sigma^2) - k/2$, and $\Delta R_{\rm st}/R_{\rm st} = 2V_{\rm W}/\sigma^2 = k^2\psi^2/\sigma^4 - k/\sigma^2$.

The Witten Hamiltonian has spectrum $E_n = nk$ with eigenstates ϕ_n.

Residual for Witten eigenstates

For $\Psi = \phi_n$ (an eigenstate of $H_{\rm W}$), $|\Psi| = |\phi_n|$, and:

$$\delta U_{\rm rel}[\phi_n] = \sigma^2 \left(\frac{\Delta|\phi_n|}{|\phi_n|} - \frac{k^2\psi^2}{\sigma^4} + \frac{k}{\sigma^2} \right) \tag{19}$$

For the ground state $\phi_0 \propto e^{-k\psi^2/(2\sigma^2)} = R_{\rm st}$: $\delta U_{\rm rel}[\phi_0] = 0$. ✓

For the first excited state $\phi_1 \propto \psi\, e^{-k\psi^2/(2\sigma^2)}$: $|\phi_1| = |\psi|\,\mathcal{N}\, e^{-k\psi^2/(2\sigma^2)}$, and one computes (away from $\psi = 0$):

$$\frac{\Delta|\phi_1|}{|\phi_1|} = \frac{k^2\psi^2}{\sigma^4} - \frac{3k}{\sigma^2} + \frac{2}{\psi^2} \tag{20}$$

So: $\delta U_{\rm rel}[\phi_1] = \sigma^2(2/\psi^2 - 2k/\sigma^2) = 2\sigma^2/\psi^2 - 2k$.

The residual diverges as $\psi \to 0$ (where $|\phi_1|$ has a node) and is negative for large $|\psi|$. The nodal divergence is expected whenever $\Delta|\Psi|/|\Psi|$ is evaluated at zeros of $|\Psi|$; the expression is understood on regions where $|\Psi| > 0$, and nodal sets require separate treatment. The residual is nonzero for excited eigenstates — confirming that only the ground state lies exactly in the Schrödinger regime.

Residual for a shifted Gaussian

Consider a Gaussian shifted from equilibrium: $|\Psi| = \mathcal{N}\, e^{-k(\psi-a)^2/(2\sigma^2)}$ for some displacement $a > 0$. Then:

$$\frac{\Delta|\Psi|}{|\Psi|} = \frac{k^2(\psi-a)^2}{\sigma^4} - \frac{k}{\sigma^2} \tag{21}$$

The residual is:

$$\delta U_{\rm rel} = \sigma^2 \left(\frac{k^2(\psi-a)^2}{\sigma^4} - \frac{k}{\sigma^2} - \frac{k^2\psi^2}{\sigma^4} + \frac{k}{\sigma^2} \right) = \frac{k^2}{\sigma^2}\left[(\psi-a)^2 - \psi^2\right] = \frac{k^2(-2a\psi + a^2)}{\sigma^2} \tag{22}$$

This is a linear function of ψ (plus a constant) — nonzero, but finite and smooth. The residual strength grows linearly with displacement a, confirming that near-equilibrium perturbations produce small residuals.

Regime summary

8 Interpretation

The closure residual $\delta U_{\rm rel}[\Psi]$ has been shown to be:

Table 3: Residual behaviour for different states in the quadratic model.

State	δU_{rel}	Regime
Ground state ϕ_0	0	Exact Schrödinger
Shifted Gaussian (small a)	$O(a)$	Linear Schrödinger
First excited state ϕ_1	Diverges at nodes	Nonlinear closure
Highly localised state	Large	Nonlinear closure

- zero at equilibrium (Proposition 2),
- phase-invariant (Proposition 3),
- local and modulus-dependent (Proposition 4),
- small near equilibrium (Proposition 5),
- not removable by phase, unitary, or equilibrium-factorisation transformations (Section 5),
- probability-conserving (Proposition 7).

These properties establish the residual as a mathematically well-defined, robust, nontrivial remainder relative to linear Schrödinger dynamics. It encodes the curvature mismatch between the actual amplitude profile and the equilibrium amplitude profile.

Remark 18 (On the role of the residual)**.** *The residual should not be interpreted as a failure to recover Schrödinger dynamics. Rather, it is the exact remainder that distinguishes the full nonlinear closure dynamics from its equilibrium tangent limit. Its presence is therefore expected: it encodes the deviation of the system from the equilibrium regime in which linear quantum behaviour is recovered.*

What the residual is not

- It is not a numerical artefact: it is an exact consequence of the closure-Nelson pairing.
- It is not a gauge-removable quantity: it survives all standard transformations.
- It is not a defect of the construction: it is the closure-sector remainder after linear quantum behaviour has been recovered.

What the residual may be

Three interpretive possibilities remain open:

1. **A physical nonlinear correction.** The residual may represent a genuine departure from linear quantum mechanics in regimes far from closure equilibrium.
2. **An effective-theory artefact.** The residual may be an artefact of the closure approximation that would vanish in a more complete framework.

3. **A structurally meaningful quantity.** The residual may encode physical information about the closure landscape that is not captured by the linearised Schrödinger dynamics.

Determining which of these holds requires further analysis — in particular, whether the residual produces observable consequences or can be constrained by known experimental bounds. This is deferred to subsequent work.

9 Discussion and Limitations

9.1 Exact results

- The equilibrium-centred normal form (3) is exact.
- The structural properties (Propositions 2–6) are proved.
- Norm conservation (Proposition 7) holds for the full nonlinear equation.
- The gauge non-removability results (Propositions 12–13) are proved.
- The residual for the quadratic example is computed explicitly.

9.2 Regime-specific results

- The near-equilibrium scaling $\delta U_{\mathrm{rel}} = O(\varepsilon)$ (Proposition 5) quantifies smallness.
- The residual-strength parameter Ξ classifies regimes but is defined pointwise; a global (integral) version would be more robust.

9.3 Open questions

- Does a modified energy functional (incorporating the residual) exist that is conserved?
- Can the residual be bounded in terms of a distance from equilibrium (e.g. relative entropy)?
- Does the nonlinear equation have special solutions not present in the linear theory?
- Can the residual be constrained by existing experimental bounds on nonlinear quantum mechanics?

10 Conclusion

Paper XIX isolates and classifies the closure residual — the sole remaining discrepancy within the closure-Nelson formulation between the paired-process dynamics and the linear Schrödinger equation.

What is established. The residual $\delta U_{\mathrm{rel}}[\Psi] = \sigma^2(\Delta|\Psi|/|\Psi| - \Delta R_{\mathrm{st}}/R_{\mathrm{st}})$ is:

- exactly zero at equilibrium,

- perturbatively small near equilibrium,
- phase-invariant, local, and modulus-dependent,
- not removable by standard gauge transformations,
- probability-conserving in all regimes.

What this means for the programme. Paper XVIII solved the kinetic problem: the imaginary kinetic term emerges from the forward/backward pairing. Paper XIX classifies the remaining nonlinear residual and shows it is a robust, non-trivial structural quantity. The dynamics is approximately linear Schrödinger in the equilibrium-centred regime ($\Xi \ll 1$) and genuinely nonlinear when the residual is not negligible.

Programme arc.

- Papers XII–XIV: dissipative closure dynamics,
- Paper XV: phase and interference,
- Paper XVI: evolution equation and structural gap,
- Paper XVII: dissipative obstruction theorem,
- Paper XVIII: Nelson pairing and near-equilibrium Schrödinger recovery,
- Paper XIX: classification of the closure residual.

The next stage is to determine the physical status of the residual: whether it is an observable nonlinear correction, an effective-theory artefact, or a structural signature of the closure landscape. That determination will decide whether the closure framework extends, modifies, or exactly recovers standard quantum mechanics.

References

Lee Smart. Quantisation as stochastic closure dynamics in φ-structured geometry. 2026a. Paper XIV. Preprint, Institute of Vibrational Field Dynamics.

Lee Smart. Nelson pairing for closure dynamics: Schrödinger-type evolution from forward and backward closure processes. 2026b. Paper XVIII. Preprint, Institute of Vibrational Field Dynamics.